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Increased accessibility of synthesized speech by replacement of difficulty to understand words

Abstract

A method, computer program, and computer system are provided for improving accessibility to computer reader tools. Data corresponding to one or more words to be displayed to a user is received. The received data is converted into synthesized speech. One or more words exceeding an understanding threshold value are identified from the synthesized speech. The understanding threshold value corresponds to a probability of difficulty associated with understanding the one or more words. One or more replacement words are retrieved for the one or more words exceeding the understanding threshold value. The synthesized speech is updated with the one or more replacement words.

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Background/Summary

FIELD

(1) This disclosure relates generally to field of machine learning, and more particularly to natural language processing.

BACKGROUND

(2) Speech synthesis is the artificial production of human speech using a computer. A computer system used for this purpose is called a speech computer or speech synthesizer, and such a system can be implemented in software or hardware products, or some combination of these. A text-to-speech system converts normal language text into speech or can render symbolic linguistic representations, like phonetic transcriptions, into speech.

SUMMARY

(3) Embodiments relate to a method, system, and computer readable medium for improving accessibility to computer reader tools. According to one aspect, a method for improving accessibility to computer reader tools is provided. The method may include receiving data corresponding to one or more words to be displayed to a user. The received data is converted into synthesized speech. One or more words exceeding an understanding threshold value are identified from the synthesized speech. The understanding threshold value corresponds to a probability of difficulty associated with understanding the one or more words. One or more replacement words are retrieved for the one or more words exceeding the understanding threshold value. The synthesized speech is updated with the one or more replacement words.

(4) According to another aspect, a computer system for improving accessibility to computer reader tools is provided. The computer system may include one or more processors, one or more computer-readable memories, one or more computer-readable tangible storage devices, and program instructions stored on at least one of the one or more storage devices for execution by at least one of the one or more processors via at least one of the one or more memories, whereby the computer system is capable of performing a method. The method may include receiving data corresponding to one or more words to be displayed to a user. The received data is converted into synthesized speech. One or more words exceeding an understanding threshold value are identified from the synthesized speech. The understanding threshold value corresponds to a probability of difficulty associated with understanding the one or more words. One or more replacement words are retrieved for the one or more words exceeding the understanding threshold value. The synthesized speech is updated with the one or more replacement words.

(5) According to yet another aspect, a computer readable medium for improving accessibility to computer reader tools is provided. The computer readable medium may include one or more computer-readable storage devices and program instructions stored on at least one of the one or more tangible storage devices, the program instructions executable by a processor. The program instructions are executable by a processor for performing a method that may accordingly include receiving data corresponding to one or more words to be displayed to a user. The received data is converted into synthesized speech. One or more words exceeding an understanding threshold value are identified from the synthesized speech. The understanding threshold value corresponds to a probability of difficulty associated with understanding the one or more words. One or more replacement words are retrieved for the one or more words exceeding the understanding threshold value. The synthesized speech is updated with the one or more replacement words.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) These and other objects, features and advantages will become apparent from the following detailed description of illustrative embodiments, which is to be read in connection with the accompanying drawings. The various features of the drawings are not to scale as the illustrations are for clarity in facilitating the understanding of one skilled in the art in conjunction with the detailed description. In the drawings:
- (2) FIG. 1 illustrates a networked computer environment according to at least one embodiment;
- (3) FIG. 2 is a block diagram of a system for improving accessibility to computer reader tools, according to at least one embodiment;
- (4) FIG. 3 is an operational flowchart illustrating the steps carried out by a program that performs improving accessibility to computer reader tools, according to at least one embodiment;
- (5) FIG. 4 is a block diagram of internal and external components of computers and servers depicted in FIG. 1 according to at least one embodiment;
- (6) FIG. 5 is a block diagram of an illustrative cloud computing environment including the computer system depicted in FIG. 1, according to at least one embodiment; and
- (7) FIG. 6 is a block diagram of functional layers of the illustrative cloud computing environment of FIG. 5, according to at least one embodiment.

DETAILED DESCRIPTION

- (8) Detailed embodiments of the claimed structures and methods are disclosed herein; however, it can be understood that the disclosed embodiments are merely illustrative of the claimed structures and methods that may be embodied in various forms. Those structures and methods may, however, be embodied in many different forms and should not be construed as limited to the exemplary embodiments set forth herein. Rather, these exemplary embodiments are provided so that this disclosure will be thorough and complete and will fully convey the scope to those skilled in the art. In the description, details of well-known features and techniques may be omitted to avoid unnecessarily obscuring the presented embodiments.
- (9) Embodiments relate generally to the field of machine learning, and more particularly to natural language processing. The following described exemplary embodiments provide a system, method, and computer program to, among other things, determine whether spoken words may be difficult to understand and replace them with spoken words that are easier to understand. Therefore, some embodiments have the capacity to improve the field of computing by allowing for computers to automatically test the accessibility of screen reader systems, and improve accessibility of computer reader tools, by increasing understanding of synthesized speech generated by them.
- (10) As previously described, speech synthesis is the artificial production of human speech. A computer system used for this purpose is called a speech computer or speech synthesizer, and such a system can be implemented in software products, hardware products, or some combination of hardware and software. A text-to-speech system converts normal language text (in the form of a corpus of text in a book, magazine, computerized document, etc.) into speech or can render symbolic linguistic representations, like phonetic transcriptions, into speech.
- (11) Many people use computer reader tools to help them work with text displayed on computer systems. While there are industry standards for screen reader solutions, the outputs of screen readers may not be of a format that facilitates testing such solutions. For example, when considering the importance of the length of a given statement input into or output by a screen reader, it may be required that the length of the message should not exceed certain values based on the capabilities of a user of the system. In addition, some words are phonetically very similar to each other, and in some cases even identical, which makes it difficult without context to correctly distinguish them once they are heard despite a of lack of problem when the user can read them.

Testing such aspects is often necessary, but manually performed testing may be very time-consuming.

(12) It may be advantageous, therefore, to automate the process of testing the accessibility of computer reader-based solutions using speech recognition. The direct flow of synthesized speech to the speech recognition solution will greatly enhance the speed in which the computer reader labels can be tested. Furthermore, replacement of a human tester by the speech recognition neural network enables to produce consistent metrics of for example word similarity to others.

(13) Aspects are described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems), and computer readable media according to the various embodiments. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer readable program instructions.

(14) The following described exemplary embodiments provide a system, method and computer program that performs automatic accessibility testing based on determining whether words may be difficult to understand and replacing them with words that are easier to understand. Referring now to FIG. 1, a functional block diagram is shown of a networked computer environment illustrating an accessibility testing system **100** (hereinafter “system”) for automatic accessibility testing. It should be appreciated that FIG. 1 provides only an illustration of one implementation and does not imply any limitations with regard to the environments in which different embodiments may be implemented. Many modifications to the depicted environments may be made based on design and implementation requirements.

(15) The system **100** may include a computer **102** and a server computer **114**. The computer **102** may communicate with the server computer **114** via a communication network **110** (hereinafter “network”). The computer **102** may include a processor **104** and a software program **108** that is stored on a data storage device **106** and is enabled to interface with a user and communicate with the server computer **114**. As will be discussed below with reference to FIG. 4 the computer **102** may include internal components **800A** and external components **900A**, respectively, and the server computer **114** may include internal components **800B** and external components **900B**, respectively. The computer **102** may be, for example, a mobile device, a telephone, a personal digital assistant, a netbook, a laptop computer, a tablet computer, a desktop computer, or any type of computing devices capable of running a program, accessing a network, and accessing a database.

(16) The server computer **114** may also operate in a cloud computing service model, such as Software as a Service (SaaS), Platform as a Service (PaaS), or Infrastructure as a Service (IaaS), as discussed below with respect to FIGS. 5 and 6. The server computer **114** may also be located in a cloud computing deployment model, such as a private cloud, community cloud, public cloud, or hybrid cloud.

(17) The server computer **114**, which may be used for automatic accessibility testing, is enabled to run an Accessibility Testing Program **116** (hereinafter “program”) that may interact with a database **112**. The Accessibility Testing Program is explained in more detail below with respect to FIG. 3. In one embodiment, the computer **102** may operate as an input device including a user interface while the program **116** may run primarily on server computer **114**. In an alternative embodiment, the program **116** may run primarily on one or more computers **102** while the server computer **114** may be used for processing and storage of data used by the program **116**. It should be noted that the program **116** may be a standalone program or may be integrated into a larger accessibility testing program.

(18) It should be noted, however, that processing for the program **116** may, in some instances be shared amongst the computers **102** and the server computers **114** in any ratio. In another embodiment, the program **116** may operate on more than one computer, server computer, or some combination of computers and server computers, for example, a plurality of computers **102** communicating across the network **110** with a single server computer **114**. In another embodiment,

for example, the program **116** may operate on a plurality of server computers **114** communicating across the network **110** with a plurality of client computers. Alternatively, the program may operate on a network server communicating across the network with a server and a plurality of client computers.

(19) The network **110** may include wired connections, wireless connections, fiber optic connections, or some combination thereof. In general, the network **110** can be any combination of connections and protocols that will support communications between the computer **102** and the server computer **114**. The network **110** may include various types of networks, such as, for example, a local area network (LAN), a wide area network (WAN) such as the Internet, a telecommunication network such as the Public Switched Telephone Network (PSTN), a wireless network, a public switched network, a satellite network, a cellular network (e.g., a fifth generation (5G) network, a long-term evolution (LTE) network, a third generation (3G) network, a code division multiple access (CDMA) network, etc.), a public land mobile network (PLMN), a metropolitan area network (MAN), a private network, an ad hoc network, an intranet, a fiber optic-based network, or the like, and/or a combination of these or other types of networks.

(20) The number and arrangement of devices and networks shown in FIG. **1** are provided as an example. In practice, there may be additional devices and/or networks, fewer devices and/or networks, different devices and/or networks, or differently arranged devices and/or networks than those shown in FIG. **1**. Furthermore, two or more devices shown in FIG. **1** may be implemented within a single device, or a single device shown in FIG. **1** may be implemented as multiple, distributed devices. Additionally, or alternatively, a set of devices (e.g., one or more devices) of system **100** may perform one or more functions described as being performed by another set of devices of system **100**.

(21) Referring now to FIG. **2**, a diagram of an accessibility testing system **200** is depicted according to one or more embodiments. The accessibility testing system **200** may include, among other things, a screen reader **202**, a neural network **204**, a database **206**, and a user interface **208**. The screen reader **202** may generate files containing synthesized speech for readable content in a document, on a website, or contained within a file. Specifically, the screen reader **202** may synthesize speech based on recognizing text to be displayed through the user interface **208**. The screen reader **202** may recognize words from the text and may split the words into individual phonemes. The screen reader **202** may generate a waveform based on the identified phonemes. The length of all words to be communicated may be measured by the screen reader **202**.

(22) The synthesized speech files may be passed to the neural network **204** and/or stored in the database **206**. The neural network **204** may record confidence scores that may indicate the certainty with which it guesses all the words. Words that have low confidence scores that may fall below an understanding threshold value may be identified and the neural network **204** may determine a synonym that may be easier to distinguish. The system may generate a report indicating words that have been replaced. The report may be a text document, spreadsheet, or audio file. By calibrating the understanding threshold value, the system may be able to return true and false outputs that may facilitate use of the system in continuous integration and continuous delivery/deployment (CI/CD) solutions.

(23) The neural network **204** may perform automated tests of the accessibility of the screen reader **202**. The neural network **204** may have pre-defined thresholds based on a level of ability of an end user. The neural network **204** may update these thresholds, or a tester may have full control over adjusting the thresholds. The user may define websites, documents, or the like for testing of the screen reader **202** by the neural network **204**. The neural network **204** may isolate screen-readable content from the website, document, etc. The neural network **204** may check that the screen-readable content contains long numbers or non-standard characters and may exclude such strings from testing. The neural network **204** may replace difficult to understand words within the synthesized speech by replacing such words with synonyms from an electronic thesaurus stored

within the database **206**. The neural network **204** may also generate a report indicating which words have been replaced, in addition to an explanation indicating a probability of difficulty in understanding the words may have been replaced. The updated synthesized speech and report may be transmitted to a user through the user interface **208**.

(24) Referring now to FIG. **3**, an operational flowchart illustrating the steps of a method **300** carried out by a program that improves accessibility to computer reader tools is depicted. The method **300** may be described with the aid of the exemplary embodiments depicted in FIGS. **1** and **2**.

(25) At **302**, the method **300** may include receiving data corresponding to one or more words to be displayed to a user. The data may be a website, document, or other file containing text to be displayed. In operation, a screen reader **202** (FIG. **2**) may receive data from a database **206** (FIG. **2**) or over a communication network **110** (FIG. **1**).

(26) At **304**, the method **300** may include converting the received data into synthesized speech. Conversion of the data to the synthesized speech may exclude strings such as large numbers, proper nouns, or words with non-standard characters. In operation, the screen reader **202** (FIG. **2**) may convert the received data to synthesized speech based on identifying words within the data, dividing the words into phonemes, and generating a waveform corresponding to the phonemes. The screen reader **202** may store files containing the synthesized speech within the database **206** (FIG. **2**).

(27) At **306**, the method **300** may include identifying, from the synthesized speech, one or more words exceeding an understanding threshold value. The understanding threshold value corresponds to a probability of difficulty associated with understanding the one or more words. The words exceeding the threshold are identified based on a neural network processing the synthesized speech and identifying a confidence value associated with certainty in understanding each word within the synthesized speech. In operation, a neural network **204** (FIG. **2**) may receive the synthesized speech files from the screen reader **202** (FIG. **2**) or may retrieve the files from the database **206** (FIG. **2**). The neural network **204** may network process the synthesized speech files and identify confidence scores for each word within the synthesized speech in order to determine whether words should be replaced due to difficulty in understanding the words.

(28) At **308**, the method **300** may include retrieving one or more replacement words for the one or more words exceeding the understanding threshold value. The replacement words are retrieved from an electronic thesaurus. In operation, the neural network **204** (FIG. **2**) may retrieve replacement words from the database **206** (FIG. **2**) for replacement within the data to be displayed to the user.

(29) At **310**, the method **300** may include updating the synthesized speech with the one or more replacement words. The updated synthesized speech may be easier to understand based on replacement of homonyms (i.e., homophones or homographs) and other difficult to understand words within the data. In operation, the neural network **204** (FIG. **2**) may update the synthesized speech with the replacement words. Specifically, the neural network **204** may make a copy of the data to be displayed that includes the replacement words and may run the copy of the updated data through the screen reader **202** to generate a new waveform for the updated synthesized speech. The updated synthesized speech may be played to the user through the user interface **208** (FIG. **2**). The user interface **208** may also display a report indicating which words may have been replaced from the original data and explanations indicating why the words were replaced.

(30) It may be appreciated that FIG. **3** provides only an illustration of one implementation and does not imply any limitations with regard to how different embodiments may be implemented. Many modifications to the depicted environments may be made based on design and implementation requirements.

(31) FIG. **4** is a block diagram **400** of internal and external components of computers depicted in FIG. **1** in accordance with an illustrative embodiment. It should be appreciated that FIG. **4** provides

only an illustration of one implementation and does not imply any limitations with regard to the environments in which different embodiments may be implemented. Many modifications to the depicted environments may be made based on design and implementation requirements.

(32) Computer **102** (FIG. **1**) and server computer **114** (FIG. **1**) may include respective sets of internal components **800A,B** and external components **900A,B** illustrated in FIG. **4**. Each of the sets of internal components **800** include one or more processors **820**, one or more computer-readable RAMs **822** and one or more computer-readable ROMs **824** on one or more buses **826**, one or more operating systems **828**, and one or more computer-readable tangible storage devices **830**.

(33) Processor **820** is implemented in hardware, firmware, or a combination of hardware and software. Processor **820** is a central processing unit (CPU), a graphics processing unit (GPU), an accelerated processing unit (APU), a microprocessor, a microcontroller, a digital signal processor (DSP), a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), or another type of processing component. In some implementations, processor **820** includes one or more processors capable of being programmed to perform a function. Bus **826** includes a component that permits communication among the internal components **800A,B**.

(34) The one or more operating systems **828**, the software program **108** (FIG. **1**) and the Accessibility Testing Program **116** (FIG. **1**) on server computer **114** (FIG. **1**) are stored on one or more of the respective computer-readable tangible storage devices **830** for execution by one or more of the respective processors **820** via one or more of the respective RAMs **822** (which typically include cache memory). In the embodiment illustrated in FIG. **4**, each of the computer-readable tangible storage devices **830** is a magnetic disk storage device of an internal hard drive. Alternatively, each of the computer-readable tangible storage devices **830** is a semiconductor storage device such as ROM **824**, EPROM, flash memory, an optical disk, a magneto-optic disk, a solid state disk, a compact disc (CD), a digital versatile disc (DVD), a floppy disk, a magnetic tape, and/or another type of non-transitory computer-readable tangible storage device that can store a computer program and digital information.

(35) Each set of internal components **800A,B** also includes a RAY drive or interface **832** to read from and write to one or more portable computer-readable tangible storage devices **936** such as a CD-ROM, DVD, memory stick, magnetic tape, magnetic disk, optical disk or semiconductor storage device. A software program, such as the software program **108** (FIG. **1**) and the Accessibility Testing Program **116** (FIG. **1**) can be stored on one or more of the respective portable computer-readable tangible storage devices **936**, read via the respective R/W drive or interface **832** and loaded into the respective hard drive **830**.

(36) Each set of internal components **800A,B** also includes network adapters or interfaces **836** such as a TCP/IP adapter cards; wireless Wi-Fi interface cards; or 3G, 4G, or 5G wireless interface cards or other wired or wireless communication links. The software program **108** (FIG. **1**) and the Accessibility Testing Program **116** (FIG. **1**) on the server computer **114** (FIG. **1**) can be downloaded to the computer **102** (FIG. **1**) and server computer **114** from an external computer via a network (for example, the Internet, a local area network or other, wide area network) and respective network adapters or interfaces **836**. From the network adapters or interfaces **836**, the software program **108** and the Accessibility Testing Program **116** on the server computer **114** are loaded into the respective hard drive **830**. The network may comprise copper wires, optical fibers, wireless transmission, routers, firewalls, switches, gateway computers and/or edge servers.

(37) Each of the sets of external components **900A,B** can include a computer display monitor **920**, a keyboard **930**, and a computer mouse **934**. External components **900A,B** can also include touch screens, virtual keyboards, touch pads, pointing devices, and other human interface devices. Each of the sets of internal components **800A,B** also includes device drivers **840** to interface to computer display monitor **920**, keyboard **930** and computer mouse **934**. The device drivers **840**, R/W drive or interface **832** and network adapter or interface **836** comprise hardware and software (stored in storage device **830** and/or ROM **824**).

(38) It is understood in advance that although this disclosure includes a detailed description on cloud computing, implementation of the teachings recited herein are not limited to a cloud computing environment. Rather, some embodiments are capable of being implemented in conjunction with any other type of computing environment now known or later developed.

(39) Cloud computing is a model of service delivery for enabling convenient, on-demand network access to a shared pool of configurable computing resources (e.g. networks, network bandwidth, servers, processing, memory, storage, applications, virtual machines, and services) that can be rapidly provisioned and released with minimal management effort or interaction with a provider of the service. This cloud model may include at least five characteristics, at least three service models, and at least four deployment models.

(40) Characteristics are as follows:

(41) On-demand self-service: a cloud consumer can unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with the service's provider.

(42) Broad network access: capabilities are available over a network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, laptops, and PDAs).

(43) Resource pooling: the provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to demand. There is a sense of location independence in that the consumer generally has no control or knowledge over the exact location of the provided resources but may be able to specify location at a higher level of abstraction (e.g., country, state, or datacenter).

(44) Rapid elasticity: capabilities can be rapidly and elastically provisioned, in some cases automatically, to quickly scale out and be rapidly released to quickly scale in. To the consumer, the capabilities available for provisioning often appear to be unlimited and can be purchased in any quantity at any time.

(45) Measured service: cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts). Resource usage can be monitored, controlled, and reported providing transparency for both the provider and consumer of the utilized service.

(46) Service Models are as follows:

(47) Software as a Service (SaaS): the capability provided to the consumer is to use the provider's applications running on a cloud infrastructure. The applications are accessible from various client devices through a thin client interface such as a web browser (e.g., web-based e-mail). The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, storage, or even individual application capabilities, with the possible exception of limited user-specific application configuration settings.

(48) Platform as a Service (PaaS): the capability provided to the consumer is to deploy onto the cloud infrastructure consumer-created or acquired applications created using programming languages and tools supported by the provider. The consumer does not manage or control the underlying cloud infrastructure including networks, servers, operating systems, or storage, but has control over the deployed applications and possibly application hosting environment configurations.

(49) Infrastructure as a Service (IaaS): the capability provided to the consumer is to provision processing, storage, networks, and other fundamental computing resources where the consumer is able to deploy and run arbitrary software, which can include operating systems and applications. The consumer does not manage or control the underlying cloud infrastructure but has control over operating systems, storage, deployed applications, and possibly limited control of select networking components (e.g., host firewalls).

(50) Deployment Models are as follows:

(51) Private cloud: the cloud infrastructure is operated solely for an organization. It may be managed by the organization or a third party and may exist on-premises or off-premises.

(52) Community cloud: the cloud infrastructure is shared by several organizations and supports a specific community that has shared concerns (e.g., mission, security requirements, policy, and compliance considerations). It may be managed by the organizations or a third party and may exist on-premises or off-premises.

(53) Public cloud: the cloud infrastructure is made available to the general public or a large industry group and is owned by an organization selling cloud services.

(54) Hybrid cloud: the cloud infrastructure is a composition of two or more clouds (private, community, or public) that remain unique entities but are bound together by standardized or proprietary technology that enables data and application portability (e.g., cloud bursting for load-balancing between clouds).

(55) A cloud computing environment is service oriented with a focus on statelessness, low coupling, modularity, and semantic interoperability. At the heart of cloud computing is an infrastructure comprising a network of interconnected nodes.

(56) Referring to FIG. 5, illustrative cloud computing environment **500** is depicted. As shown, cloud computing environment **500** comprises one or more cloud computing nodes **10** with which local computing devices used by cloud consumers, such as, for example, personal digital assistant (PDA) or cellular telephone **54A**, desktop computer **54B**, laptop computer **54C**, and/or automobile computer system **54N** may communicate. Cloud computing nodes **10** may communicate with one another. They may be grouped (not shown) physically or virtually, in one or more networks, such as Private, Community, Public, or Hybrid clouds as described hereinabove, or a combination thereof. This allows cloud computing environment **500** to offer infrastructure, platforms and/or software as services for which a cloud consumer does not need to maintain resources on a local computing device. It is understood that the types of computing devices **54A-N** shown in FIG. 5 are intended to be illustrative only and that cloud computing nodes **10** and cloud computing environment **500** can communicate with any type of computerized device over any type of network and/or network addressable connection (e.g., using a web browser).

(57) Referring to FIG. 6, a set of functional abstraction layers **600** provided by cloud computing environment **500** (FIG. 5) is shown. It should be understood in advance that the components, layers, and functions shown in FIG. 6 are intended to be illustrative only and embodiments are not limited thereto. As depicted, the following layers and corresponding functions are provided:

(58) Hardware and software layer **60** includes hardware and software components. Examples of hardware components include: mainframes **61**; RISC (Reduced Instruction Set Computer) architecture based servers **62**; servers **63**; blade servers **64**; storage devices **65**; and networks and networking components **66**. In some embodiments, software components include network application server software **67** and database software **68**.

(59) Virtualization layer **70** provides an abstraction layer from which the following examples of virtual entities may be provided: virtual servers **71**; virtual storage **72**; virtual networks **73**, including virtual private networks; virtual applications and operating systems **74**; and virtual clients **75**.

(60) In one example, management layer **80** may provide the functions described below. Resource provisioning **81** provides dynamic procurement of computing resources and other resources that are utilized to perform tasks within the cloud computing environment. Metering and Pricing **82** provide cost tracking as resources are utilized within the cloud computing environment, and billing or invoicing for consumption of these resources. In one example, these resources may comprise application software licenses. Security provides identity verification for cloud consumers and tasks, as well as protection for data and other resources. User portal **83** provides access to the cloud computing environment for consumers and system administrators. Service level management **84**

provides cloud computing resource allocation and management such that required service levels are met. Service Level Agreement (SLA) planning and fulfillment **85** provide pre-arrangement for, and procurement of, cloud computing resources for which a future requirement is anticipated in accordance with an SLA.

(61) Workloads layer **90** provides examples of functionality for which the cloud computing environment may be utilized. Examples of workloads and functions which may be provided from this layer include: mapping and navigation **91**; software development and lifecycle management **92**; virtual classroom education delivery **93**; data analytics processing **94**; transaction processing **95**; and Accessibility Testing **96**. Accessibility Testing **96** may determine whether words may be difficult to understand and replace them with words that are easier to understand.

(62) Some embodiments may relate to a system, a method, and/or a computer readable medium at any possible technical detail level of integration. The computer readable medium may include a computer-readable non-transitory storage medium (or media) having computer readable program instructions thereon for causing a processor to carry out operations.

(63) The computer readable storage medium can be a tangible device that can retain and store instructions for use by an instruction execution device. The computer readable storage medium may be, for example, but is not limited to, an electronic storage device, a magnetic storage device, an optical storage device, an electromagnetic storage device, a semiconductor storage device, or any suitable combination of the foregoing. A non-exhaustive list of more specific examples of the computer readable storage medium includes the following: a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a static random access memory (SRAM), a portable compact disc read-only memory (CD-ROM), a digital versatile disk (DVD), a memory stick, a floppy disk, a mechanically encoded device such as punch-cards or raised structures in a groove having instructions recorded thereon, and any suitable combination of the foregoing. A computer readable storage medium, as used herein, is not to be construed as being transitory signals per se, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide or other transmission media (e.g., light pulses passing through a fiber-optic cable), or electrical signals transmitted through a wire.

(64) Computer readable program instructions described herein can be downloaded to respective computing/processing devices from a computer readable storage medium or to an external computer or external storage device via a network, for example, the Internet, a local area network, a wide area network and/or a wireless network. The network may comprise copper transmission cables, optical transmission fibers, wireless transmission, routers, firewalls, switches, gateway computers and/or edge servers. A network adapter card or network interface in each computing/processing device receives computer readable program instructions from the network and forwards the computer readable program instructions for storage in a computer readable storage medium within the respective computing/processing device.

(65) Computer readable program code/instructions for carrying out operations may be assembler instructions, instruction-set-architecture (ISA) instructions, machine instructions, machine dependent instructions, microcode, firmware instructions, state-setting data, configuration data for integrated circuitry, or either source code or object code written in any combination of one or more programming languages, including an object oriented programming language such as Smalltalk, C++, or the like, and procedural programming languages, such as the “C” programming language or similar programming languages. The computer readable program instructions may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an

Internet Service Provider). In some embodiments, electronic circuitry including, for example, programmable logic circuitry, field-programmable gate arrays (FPGA), or programmable logic arrays (PLA) may execute the computer readable program instructions by utilizing state information of the computer readable program instructions to personalize the electronic circuitry, in order to perform aspects or operations.

(66) These computer readable program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks. These computer readable program instructions may also be stored in a computer readable storage medium that can direct a computer, a programmable data processing apparatus, and/or other devices to function in a particular manner, such that the computer readable storage medium having instructions stored therein comprises an article of manufacture including instructions which implement aspects of the function/act specified in the flowchart and/or block diagram block or blocks.

(67) The computer readable program instructions may also be loaded onto a computer, other programmable data processing apparatus, or other device to cause a series of operational steps to be performed on the computer, other programmable apparatus or other device to produce a computer implemented process, such that the instructions which execute on the computer, other programmable apparatus, or other device implement the functions/acts specified in the flowchart and/or block diagram block or blocks.

(68) The flowchart and block diagrams in the Figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer readable media according to various embodiments. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of instructions, which comprises one or more executable instructions for implementing the specified logical function(s). The method, computer system, and computer readable medium may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in the Figures. In some alternative implementations, the functions noted in the blocks may occur out of the order noted in the Figures. For example, two blocks shown in succession may, in fact, be executed concurrently or substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts or carry out combinations of special purpose hardware and computer instructions.

(69) It will be apparent that systems and/or methods, described herein, may be implemented in different forms of hardware, firmware, or a combination of hardware and software. The actual specialized control hardware or software code used to implement these systems and/or methods is not limiting of the implementations. Thus, the operation and behavior of the systems and/or methods were described herein without reference to specific software code—it being understood that software and hardware may be designed to implement the systems and/or methods based on the description herein.

(70) No element, act, or instruction used herein should be construed as critical or essential unless explicitly described as such. Also, as used herein, the articles “a” and “an” are intended to include one or more items, and may be used interchangeably with “one or more.” Furthermore, as used herein, the term “set” is intended to include one or more items (e.g., related items, unrelated items, a combination of related and unrelated items, etc.), and may be used interchangeably with “one or more.” Where only one item is intended, the term “one” or similar language is used. Also, as used herein, the terms “has,” “have,” “having,” or the like are intended to be open-ended terms. Further,

the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise.

(71) The descriptions of the various aspects and embodiments have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Even though combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of possible implementations. In fact, many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. Although each dependent claim listed below may directly depend on only one claim, the disclosure of possible implementations includes each dependent claim in combination with every other claim in the claim set. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

Claims

1. A method of improving accessibility to computer reader tools, the method executable by a processor and comprising: receiving data corresponding to one or more words to be displayed to a user; filtering the received data to remove numbers, proper nouns, and non-standard characters; converting the filtered data into synthesized speech; identifying, from the synthesized speech, one or more words exceeding an understanding threshold value, the understanding threshold value corresponding to a probability of difficulty associated with understanding the one or more words, wherein the identifying comprises a neural network processing the synthesized speech and identifying a confidence score associated with each word within the synthesized speech; retrieving one or more replacement words for the one or more words exceeding the understanding threshold value, wherein the one or more replacement words are synonyms retrieved from an electronic thesaurus; updating the synthesized speech with the one or more replacement words, wherein the updating comprises generating a new waveform of updated synthesized speech with the one or more replacement words; generating a report indicating the one or more words and the one or more replacement words, wherein the report comprises an audio file; updating, via the neural network, the understanding threshold value based on the report; and playing the updated synthesized speech and the report to the user.
2. The method of claim 1, wherein the report includes an explanation corresponding to the probability of difficulty associated with understanding the one or more words.
3. The method of claim 2, wherein the explanation corresponding to the probability of difficulty associated with understanding the one or more words includes the one or more words being homophones or homographs of other words.
4. A computer system for improving accessibility to computer reader tools, the computer system comprising: one or more computer-readable non-transitory storage media configured to store computer program code; and one or more computer processors configured to access said computer program code and operate as instructed by said computer program code, said computer program code including: receiving code configured to cause the one or more computer processors to receive data corresponding to one or more words to be displayed to a user; filtering code configured to cause the one or more computer processors to filter the received data to remove numbers, proper nouns, and non-standard characters; converting code configured to cause the one or more computer processors to convert the filtered data into synthesized speech; identifying code configured to cause the one or more computer processors to identify, from synthesized speech, one or more words exceeding an understanding threshold value, the understanding threshold value corresponding to a probability of difficulty associated with understanding the one or more words, wherein the

identifying comprises a neural network processing the synthesized speech and identifying a confidence score associated with each word within the synthesized speech; retrieving code configured to cause the one or more computer processors to retrieve one or more replacement words for the one or more words exceeding the understanding threshold value, wherein the one or more replacement words are synonyms retrieved from an electronic thesaurus; updating code configured to cause the one or more computer processors to update the synthesized speech with the one or more replacement words, wherein the updating comprises generating a new waveform of updated synthesized speech with the one or more replacement words; generating a report indicating the one or more words and the one or more replacement words, wherein the report comprises an audio file; updating code configured to cause the one or more computer processors to update, via the neural network, the understanding threshold value based on the report; and playing code configured to cause the one or more computer processors to play the updated synthesized speech and the report to the user.

5. The computer system of claim 4, wherein the report includes an explanation corresponding to the probability of difficulty associated with understanding the one or more words.

6. The computer system of claim 5, wherein the explanation corresponding to the probability of difficulty associated with understanding the one or more words includes the one or more words being homophones or homographs of other words.

7. A non-transitory computer readable medium having stored thereon a computer program for improving accessibility to computer reader tools, the computer program configured to cause one or more computer processors to: receive data corresponding to one or more words to be displayed to a user; filter the received data to remove numbers, proper nouns, and non-standard characters; convert the filtered data into synthesized speech; identify, from synthesized speech, one or more words exceeding an understanding threshold value, the understanding threshold value corresponding to a probability of difficulty associated with understanding the one or more words, wherein the identifying comprises a neural network processing the synthesized speech and identifying a confidence score associated with each word within the synthesized speech; retrieve one or more replacement words for the one or more words exceeding the understanding threshold value, wherein the one or more replacement words are synonyms retrieved from an electronic thesaurus; update the synthesized speech with the one or more replacement words, wherein the update comprises generating a new waveform of updated synthesized speech with the one or more replacement words; generating a report indicating the one or more words and the one or more replacement words, wherein the report comprises an audio file; updating, via the neural network, the understanding threshold value based on the report; and play the updated synthesized speech and the report to the user.

8. The computer readable medium of claim 7, wherein the report includes an explanation corresponding to the probability of difficulty associated with understanding the one or more words.

9. The computer readable medium of claim 8, wherein the explanation corresponding to the probability of difficulty associated with understanding the one or more words includes the one or more words being homophones or homographs of other words.
